

```

1  FUNCTION Z_FILE_PDF_ENCRYPT.
2  *"-
3  *""Local Interface:
4  *"  IMPORTING
5  *"    REFERENCE(PDF_CONTENT) TYPE  XSTRING
6  *"    REFERENCE(OWNER_PW) TYPE  CHAR12
7  *"    REFERENCE(USER_PW) TYPE  CHAR12
8  *"    REFERENCE(ALLOW_PRINT) TYPE  XFELD OPTIONAL
9  *"    REFERENCE(ALLOW_COPY) TYPE  XFELD OPTIONAL
10 *"    REFERENCE(ALLOW_MODIFY) TYPE  XFELD OPTIONAL
11 *"    REFERENCE(UNIQUE_ID) TYPE  CHAR12 OPTIONAL
12 *"    REFERENCE(PATHNAME) TYPE  STRING OPTIONAL
13 *"    REFERENCE(DELETE_FILES) TYPE  CHAR1 OPTIONAL
14 *"  EXPORTING
15 *"    REFERENCE(PDF_ENCRYPTED) TYPE  XSTRING
16 *"    REFERENCE(PDF_SIZE) TYPE  SO_OBJ_LEN
17 *"  EXCEPTIONS
18 *"    ENCRYPT_ERROR
19 *"    ENQUEUE_ERROR
20 *"    NO_PERMISSION
21 *"-
22  data: params LIKE  SXPGCOLIST-PARAMETERS.
23  data: exec_proto type table of BTCXPM.
24
25  data:
26    fullfn1 type string,
27    fullfn2 type string,
28    fname1(30) value 'EncryptPDF_Source',
29    fname2(30) value 'EncryptPDF_Output',
30    time1 type i,
31    time2(8),
32    errflag(1),
33    unique_id1 type char12.
34
35  data:
36  estatus like EXTCMDSEXEX-STATUS,
37  exitcode LIKE  EXTCMDSEXEX-EXITCODE.
38
39  data:
40    voldir type TXW_LOCK2-VOLDIR SET,
41    filename type TXW_LOCK2-tax_file,
42    lock_ok(1).
43
44  unique_id1 = unique_id.
45  if unique_id1 is initial.
46    unique_id1 = sy-uname.
47  endif.
48
49  write sy-uzeit to time2 left-justified.
50  replace all occurrences of ':' in time2 with '_'.
51
52  concatenate fname1 '_' unique_id1 '_' time2 '.pdf' into fullfn1.
53  if pathname is not initial.
54    concatenate pathname fullfn1 into fullfn1.
55  endif.
56  condense fullfn1 no-gaps.
57  delete dataset fullfn1.
58
59  concatenate fname2 '_' unique_id1 '_' time2 '.pdf' into fullfn2.
60  if pathname is not initial.
61    concatenate pathname fullfn2 into fullfn2.

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62     endif.
63     condense fullfn2 no-gaps.
64     delete dataset fullfn2.
65
66     voldir = sy-uname.
67
68     lock_ok = 'N'.
69     do 10 times.
70
71         CALL FUNCTION 'ENQUEUE_E_TXW_FILE'
72             EXPORTING
73                 MODE_TXW_LOCK2 = 'X'
74                 VOLDIR_SET      = voldir
75                 TAX_FILE        = 'ENCRYPTPDF'
76             EXCEPTIONS
77                 FOREIGN_LOCK    = 1
78                 SYSTEM_FAILURE  = 2.
79
80     case sy-subrc.
81
82         when 0.
83             lock_ok = 'Y'.
84             exit.
85         when 1.
86             wait up to 2 seconds.
87         when others.
88             raise enqueue_error.
89             return.
90     endcase.
91
92     enddo.
93     if lock_ok ne 'Y'.
94         clear pdf_encrypted.
95         pdf_size = 0.
96         raise enqueue_error.
97         return.
98     endif.
99
100 * Output Source PDF xstring to file
101 open dataset fullfn1 for output in binary mode.
102 transfer pdf_content to fullfn1.
103 close dataset fullfn1.
104
105 * Run the file through the encryption program
106 concatenate
107     '-r' '"23D6X2337389BJ91"'      "Licence key
108     '-i' fullfn1                    "Input file
109     '-o' fullfn2                    "Output file
110     '-w' owner_pw                   "Owner password
111     '-u' user_pw                    "User password
112     '-e 128'                        "Encryption level
113 into params separated by space.
114 if not allow_print is initial.
115     concatenate params '-p' into params separated by space.
116 endif.
117 if not allow_copy is initial.
118     concatenate params '-c' into params separated by space.
119 endif.
120 if not allow_modify is initial.
121     concatenate params '-m' into params separated by space.
122 endif.

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123
124 clear errflag.
125 CALL FUNCTION 'SXPG_CALL_SYSTEM'
126     EXPORTING
127         COMMANDNAME                = 'ZPDFENCRYPT'
128         ADDITIONAL_PARAMETERS      = params
129 *      TRACE                        =
130     IMPORTING
131         STATUS                      = estatus
132         EXITCODE                   = exitcode
133     TABLES
134         EXEC_PROTOCOL              = exec_proto
135     EXCEPTIONS
136         NO_PERMISSION              = 1
137         COMMAND_NOT_FOUND          = 2
138         PARAMETERS_TOO_LONG        = 3
139         SECURITY_RISK               = 4
140         WRONG_CHECK_CALL_INTERFACE = 5
141         PROGRAM_START_ERROR         = 6
142         PROGRAM_TERMINATION_ERROR  = 7
143         X_ERROR                    = 8
144         PARAMETER_EXPECTED         = 9
145         TOO_MANY_PARAMETERS        = 10
146         ILLEGAL_COMMAND            = 11
147         OTHERS                     = 12.
148
149
150 if sy-subrc > 0.
151     if sy-subrc = 1.
152         errflag = '1'.
153     else.
154         errflag = '2'.
155     endif.
156     pdf_size = 0.
157     clear pdf_encrypted.
158 else.
159
160 * Read the encrypted file back into a xstring
161     open dataset fullfn2 for input in binary mode.
162     read dataset fullfn2 into pdf_encrypted.
163     close dataset fullfn2.
164     pdf_size = xstrlen( pdf_encrypted ).
165
166 endif.
167
168 * Delete the input and output files
169 if DELETE_FILES = 'X'.
170     delete dataset fullfn1.
171     delete dataset fullfn2.
172 endif.
173 wait up to 1 seconds.
174
175 CALL FUNCTION 'DEQUEUE_E_TXW_FILE'
176     EXPORTING
177         MODE_TXW_LOCK2 = 'X'
178         VOLDIR_SET     = voldir
179         TAX_FILE        = 'ENCRYPTPDF'
180 *      X_VOLDIR_SET    = ' '
181 *      X_TAX_FILE      = ' '
182 *      SCOPE           = '3'
183 *      SYNCHRON        = ' '

```

```
184 *      _COLLECT      = ' '
185      .
186
187      case errflag.
188          when '1'.
189              raise no_permission.
190          when '2'.
191              raise encrypt_error.
192      endcase.
193
194
195 ENDFUNCTION.
```

External calls

Function modules:	Line	Page
DEQUEUE_E_TXW_FILE..... LZFILEU02	175	3
ENQUEUE_E_TXW_FILE..... LZFILEU02	71	2
SXPG_CALL_SYSTEM..... LZFILEU02	125	3

```
1  FUNCTION-POOL zfile.                                "MESSAGE-ID ..
2
3  DATA: BEGIN OF file,
4      dirname      TYPE dirname_all1, " name of directory
5      name          TYPE filename_all1," name of entry
6      type(10)      TYPE c,           " type of entry.
7      len(8)        TYPE p,           " length in bytes.
8      owner(8)      TYPE c,           " owner of the entry.
9      mtime(6)      TYPE p,           " last mod.date, sec since 1970
10     mode(9)        TYPE c,           " like "rwx-r-x—x": prot. mode
11     useable(1)     TYPE c,
12     subrc(4)       TYPE c,
13     errno(3)       TYPE c,
14     errmsg(40)     TYPE c,
15     mod_date       TYPE d,
16     mod_time(8)    TYPE c,           " hh:mm:ss
17     seen(1)        TYPE c,
18     changed(1)     TYPE c,
19 END OF file.
20
21 *DATA: BEGIN OF file_list OCCURS 100,
22 *     dirname      TYPE dirname_all1, " name of directory
23 *     name          TYPE filename_all1," name of entry
24 *     type(10)      TYPE c,           " type of entry.
25 *     len(8)        TYPE p,           " length in bytes.
26 *     owner(8)      TYPE c,           " owner of the entry.
27 *     mtime(6)      TYPE p,           " last mod. date, sec since 1970
28 *     mode(9)        TYPE c,           " like "rwx-r-x—x": prot. mode
29 *     useable(1)     TYPE c,
30 *     subrc(4)       TYPE c,
31 *     errno(3)       TYPE c,
32 *     errmsg(40)     TYPE c,
33 *     mod_date       TYPE d,
34 *     mod_time(8)    TYPE c,           " hh:mm:ss
35 *     seen(1)        TYPE c,
36 *     changed(1)     TYPE c,
37 *     END OF file_list.
38
39 *DATA: file_list LIKE zfilelist OCCURS 100 WITH HEADER LINE.
40
41 DATA: sap_yes(1)  VALUE 'X'
42       , sap_no(1)  VALUE ' '.
43
44 DATA: errcnt(2) TYPE p VALUE 0.
```

```
1  *-----*
2  ***INCLUDE LZFILEF01 .
3  *-----*
4  *&-----*
5  *&      Form  FILENAME_USEABLE
6  *&-----*
7  *      text
8  *-----*
9  *      -->P_FILE_NAME  text
10 *      -->P_FILE_USEABLE  text
11 *-----*
12 FORM filename_useable USING a_name a_useable.
13 *=====
14     DATA l_name(75).
15
16     l_name = a_name.
17     IF l_name(4) = 'core'.
18         a_useable = sap_no.
19     ELSE.
20         a_useable = sap_yes.
21     ENDIF.
22 ENDFORM.                                "FILENAME_USEABLE
```

Subroutines:	Line	Page
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